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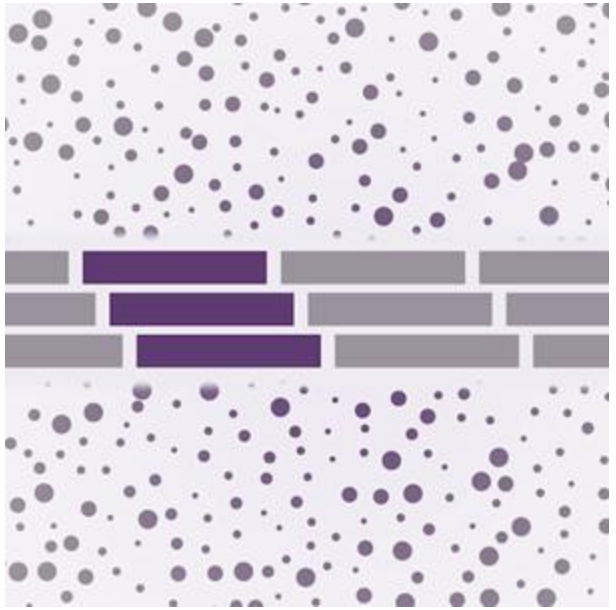
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Module #1a: Intro to Sequencing Technologies

Dr. Erin Pleasance

CBW High-Throughput Genomics

November 4-6, 2026



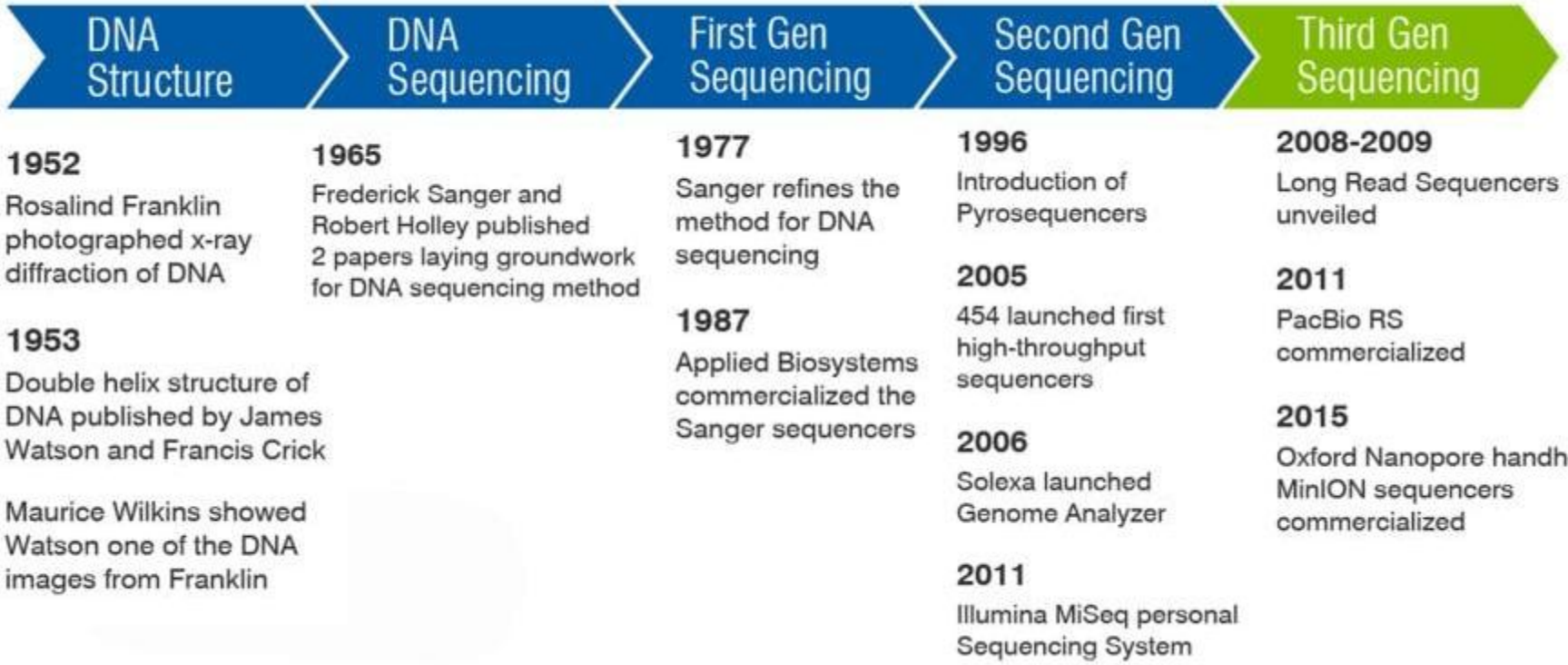
CANADA'S MICHAEL SMITH GENOME SCIENCES CENTRE

Some slides adapted from Jared Simpson and previous instructors

Learning Objectives

- Sequencing trends
- Understand the technology behind Illumina, Nanopore and PacBio sequencing
- Compare and contrast the approaches
- Be aware of newer technologies becoming available
- Understand basic sequencing reads and quality measures

DNA sequencing history



Newer technologies competing in short-read space:

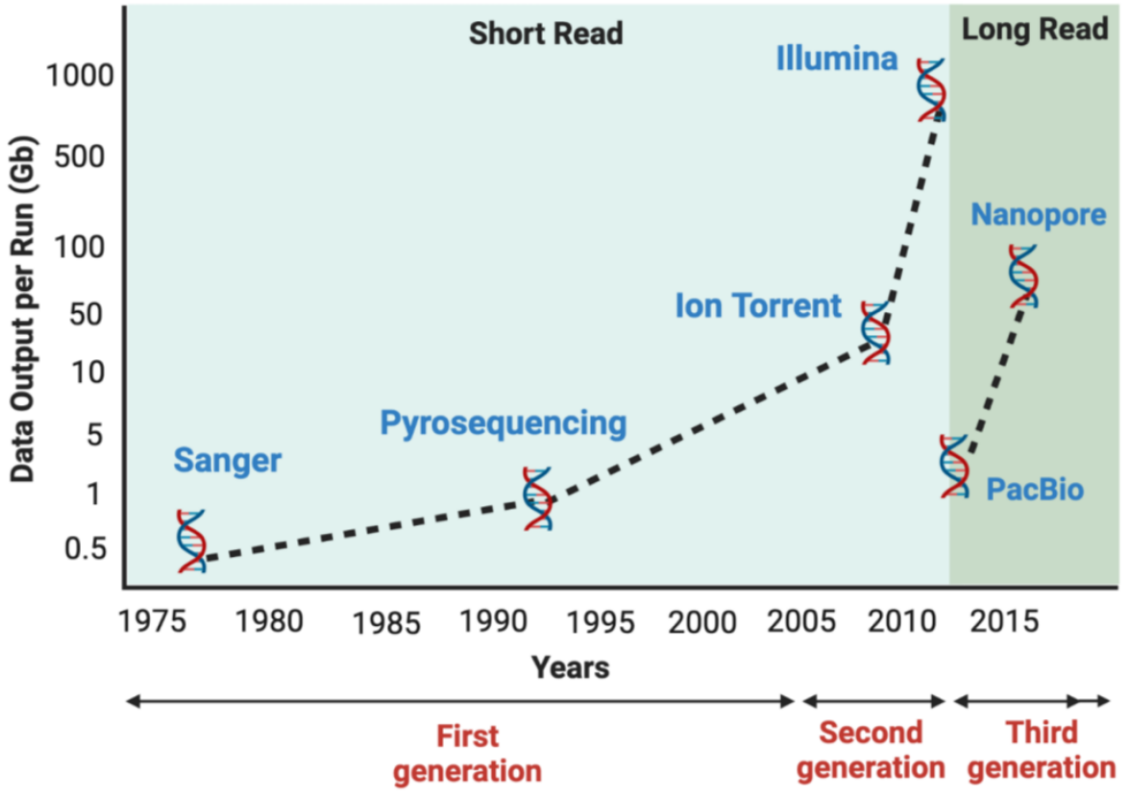
2022
AVITI (Element Biosciences)

2023
Complete Genomics

2024
Ultima

https://www.criver.com/eureka/history-dna-sequencing

Throughput and cost



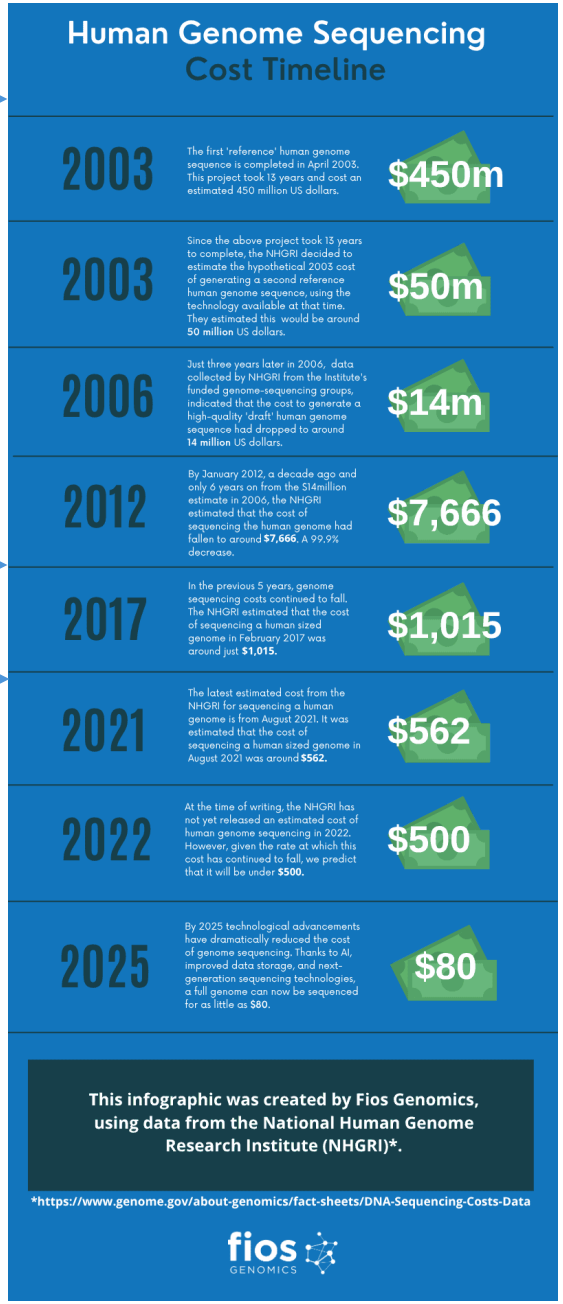
PMID: 37508427

Human genome project complete

1,000 genomes project complete

100,000 genomes project complete

All Of Us / 1+ Million Genomes

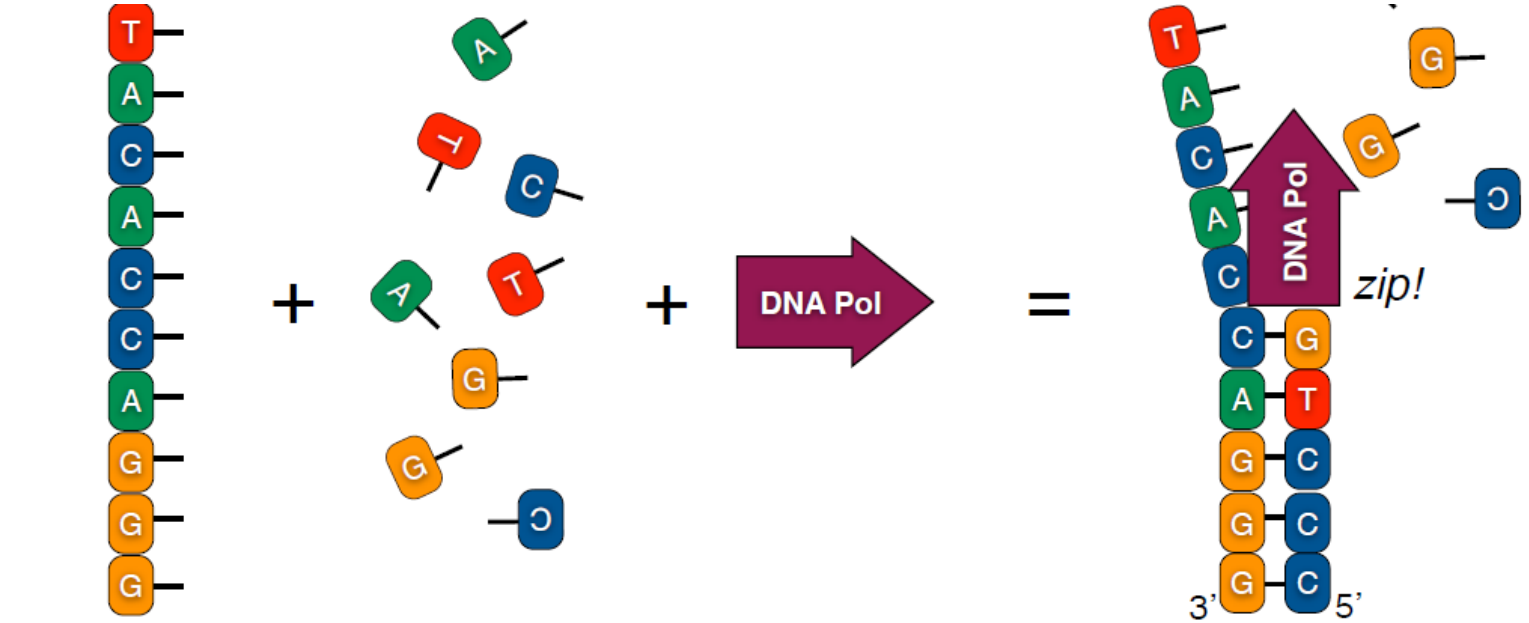


https://www.fiosgenomics.com/genomics-sequencing-data-analysis-a-cost-timeline/

Established players in sequencing

- Illumina
- Oxford Nanopore
- PacBio

DNA sequencing: DNA polymerase



Single-stranded DNA *template* | Free nucleotides | DNA polymerase | Strand synthesis

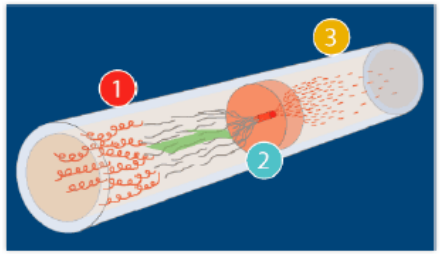
DNA polymerase moves along the template in one direction, integrating complementary nucleotides as it goes

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Illumina sequencing by synthesis

1. Take DNA sample, which includes many copies of the genome, and chop it into single-stranded fragments ("templates")

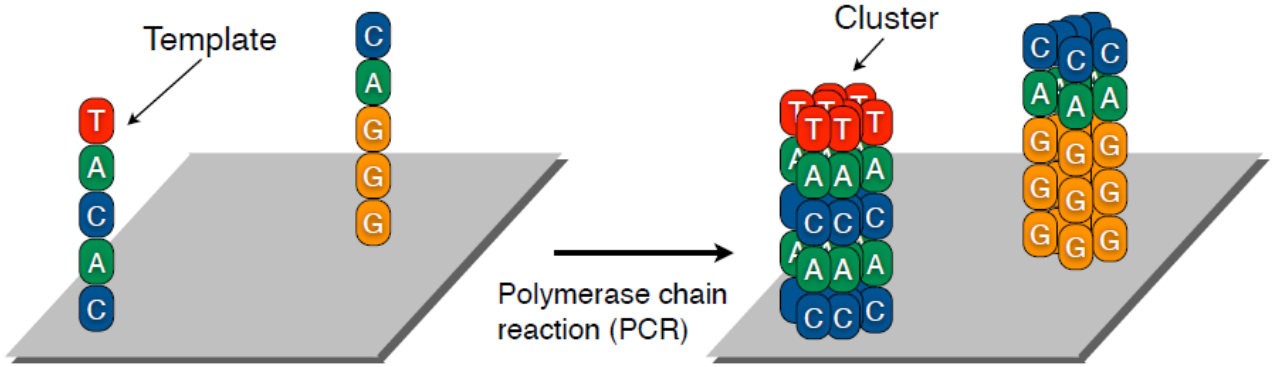
   E.g. with ultrasound waves, water-jet shearing (pictured), divalent cations



Picture: http://www.jgi.doe.gov/sequencing/education/how/how_1.html

2. Attach templates to a surface

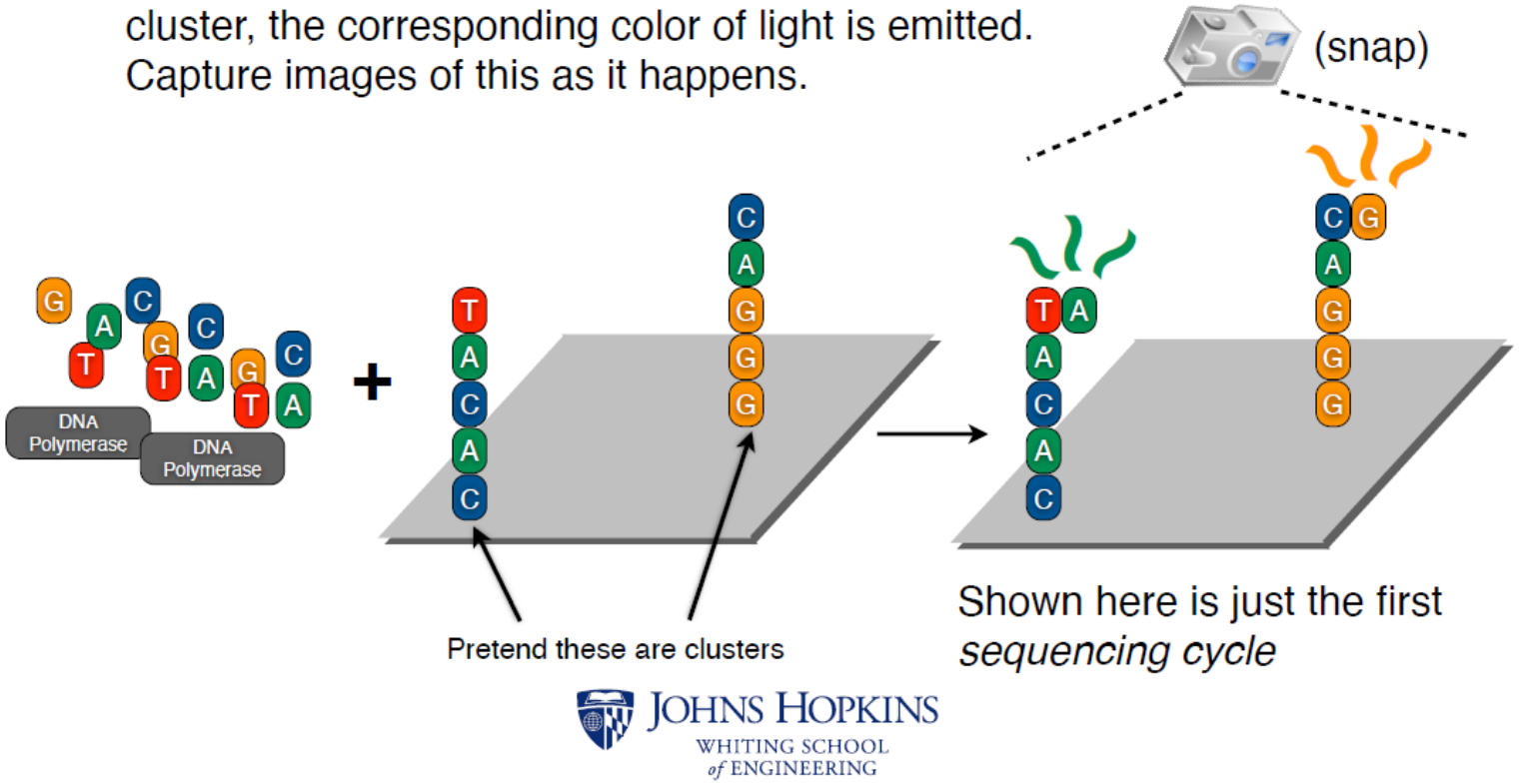
3. Make copies so that each template becomes a "cluster" of clones



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Illumina sequencing by synthesis

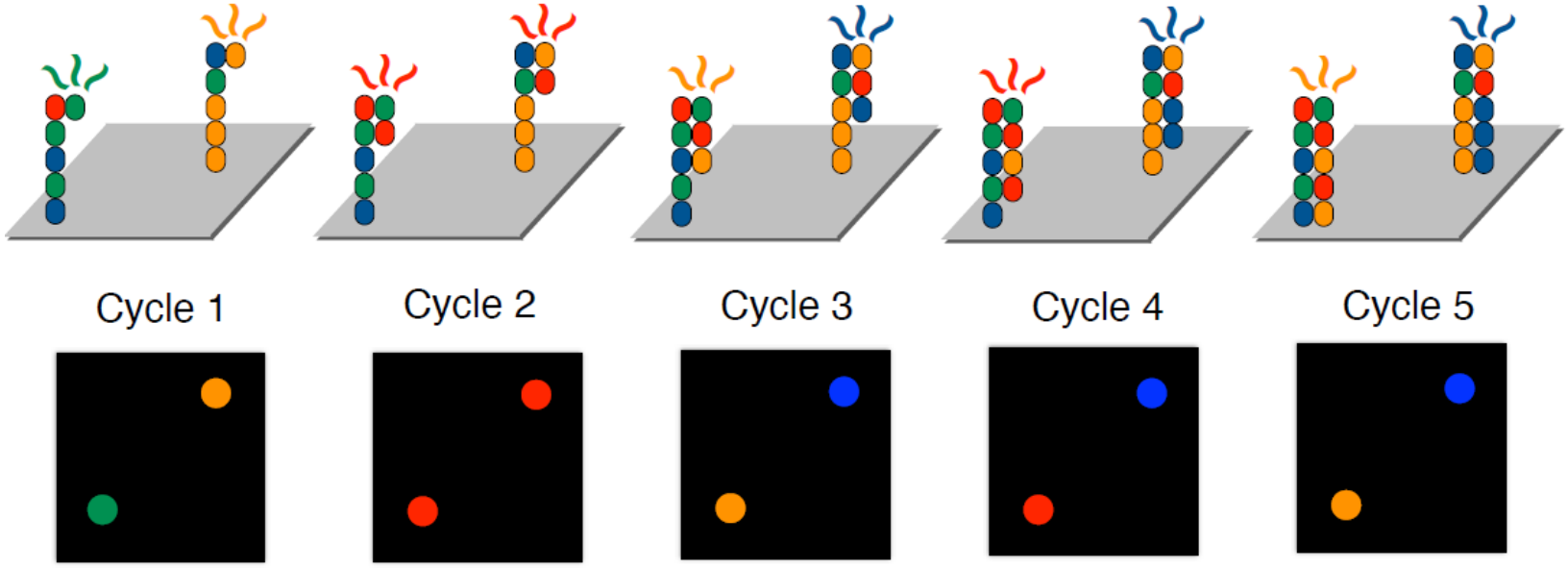
4. Repeatedly inject mixture of *color-labeled* nucleotides (A, C, G and T) and DNA polymerase. When a complementary nucleotide is added to a cluster, the corresponding color of light is emitted. Capture images of this as it happens.



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Illumina sequencing by synthesis

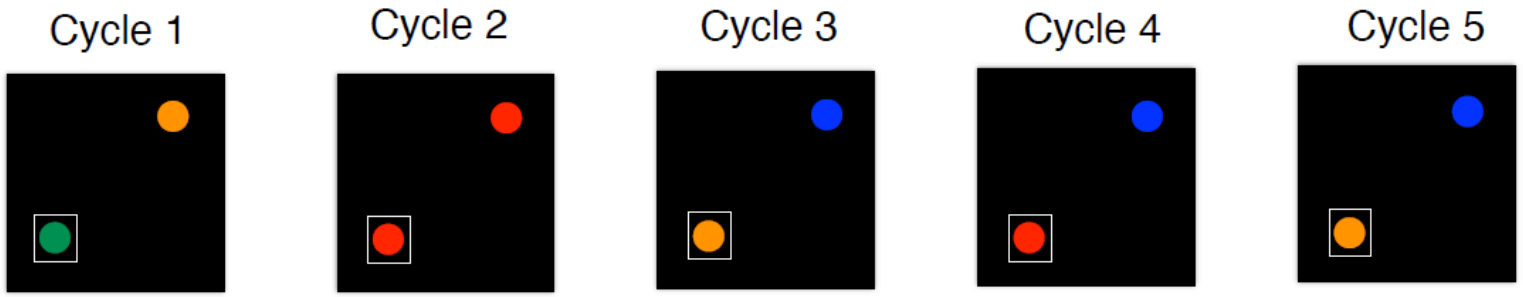
5. Line up images and, for each cluster, turn the series of light signals into corresponding series of nucleotides



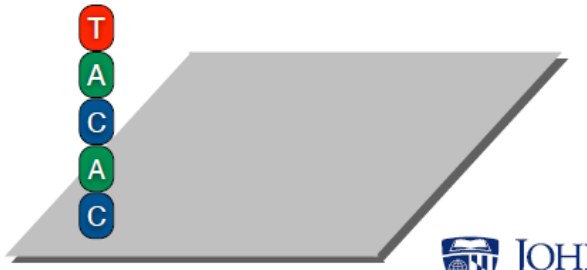
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Illumina sequencing by synthesis

5. Line up images and, for each cluster, turn the series of light signals into corresponding series of nucleotides



"Base caller" software looks at this cluster across all images and "calls" the complementary nucleotides: TACAC, corresponding to the template sequence

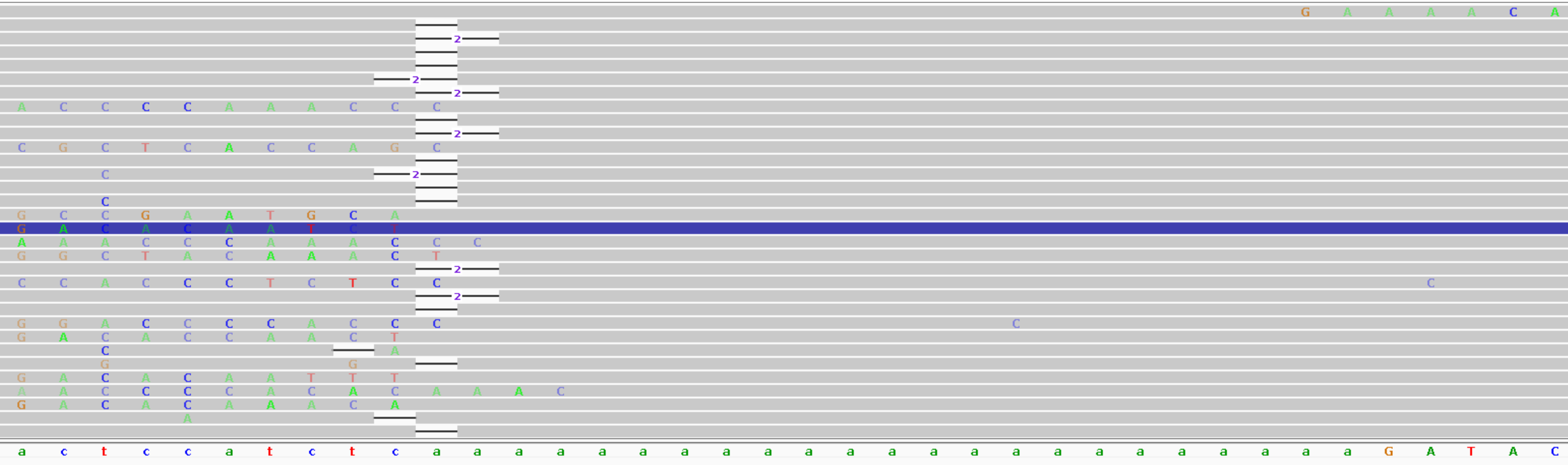


TACAC is a "sequence read," or "read." Actual reads are usually 100 or more nucleotides long.

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Homopolymers are tricky to sequence

- Polymerases can slip on long sequences of the same nucleotide
- Get lots of indels



Illumina read profile

- Read pairs facing inwards 
- Length usually 150 bp each end of pair
- Accuracy > 99.9%
- Only ACTG, no modified bases

Illumina summary

- High throughput, high accuracy, low cost
- Different platforms for different needs
- Most commonly used technology
- Read length limited to ~150 bp each pair
- No base modifications read

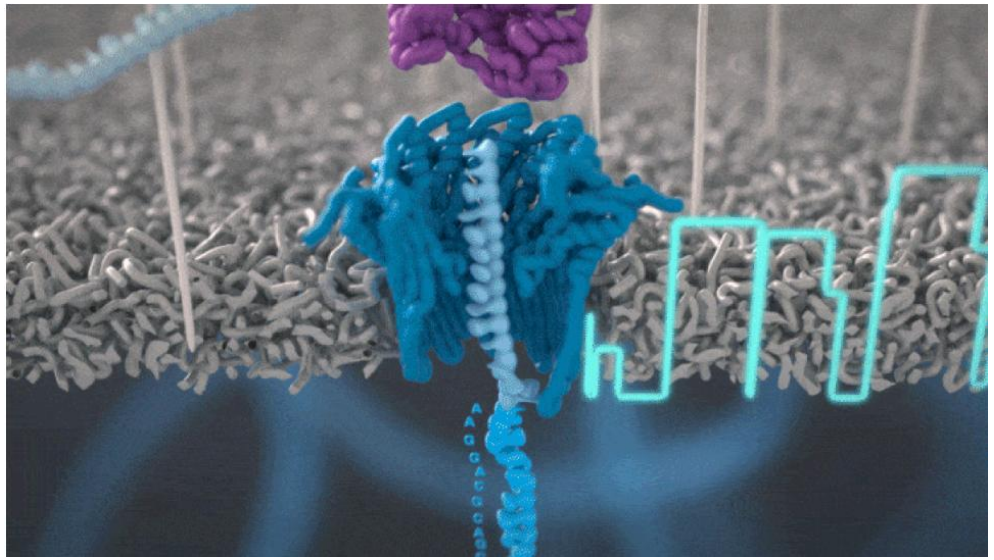


NovaSeq X Plus



MiSeq i100

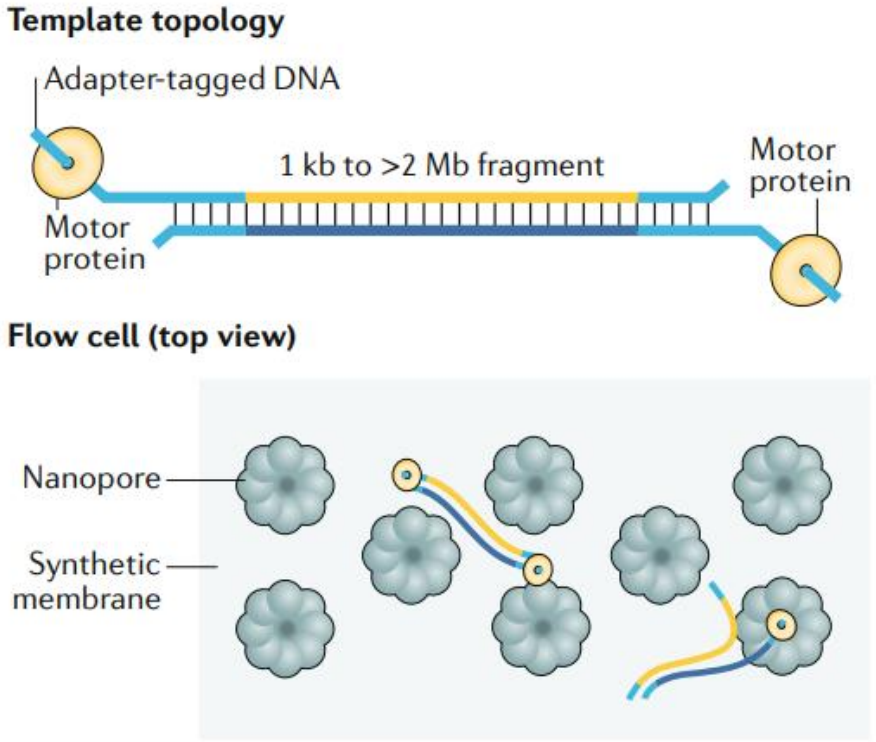
Oxford Nanopore (ONT) sequencing



- Unique nanopore technology
- Not sequencing by synthesis

https://nanoporetech.com/platform/technology

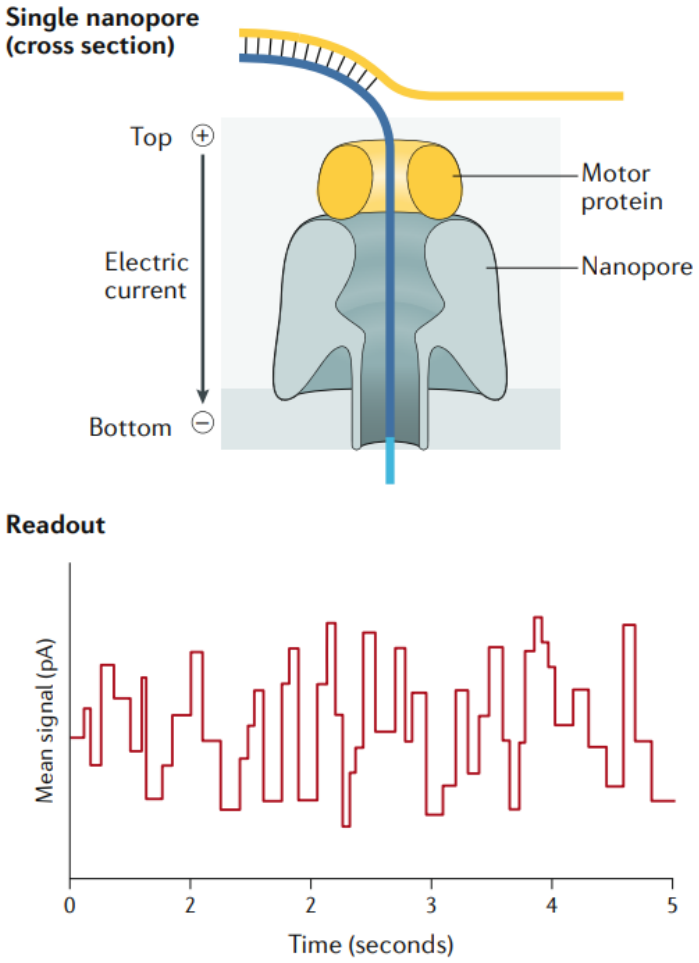
Oxford Nanopore sequencing



- Native DNA or RNA
- No amplification
- Size selection not necessary

PMID: 32504078

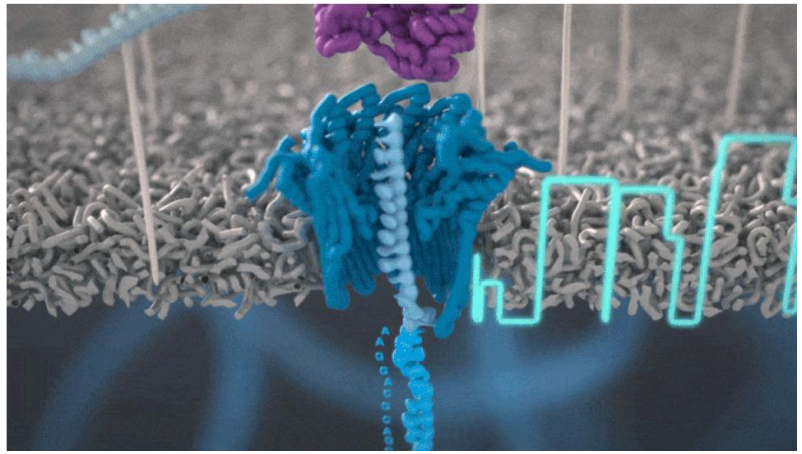
Oxford Nanopore sequencing

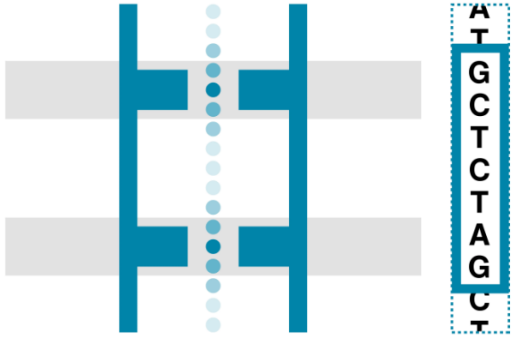


- Motor protein moves DNA through the pore
- Detect changes in electrical signal depending on sequence in the pore (“squiggle”)

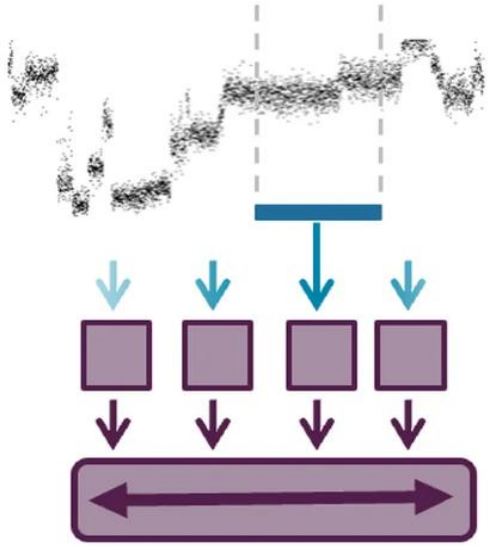
PMID: 32504078

Oxford Nanopore sequencing



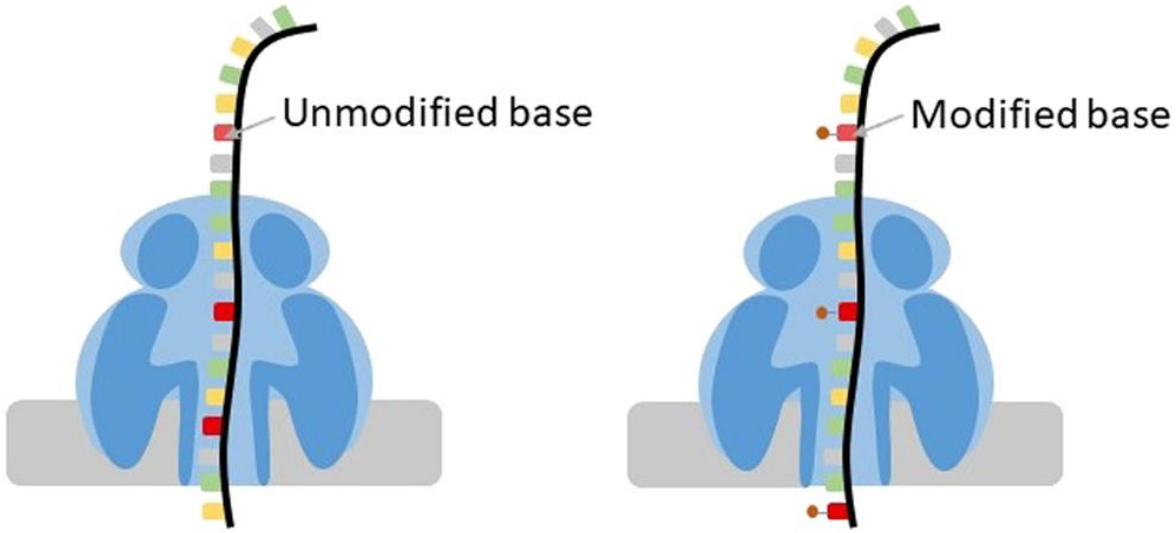


- Signal depends on multiple bases
- Basecaller decodes

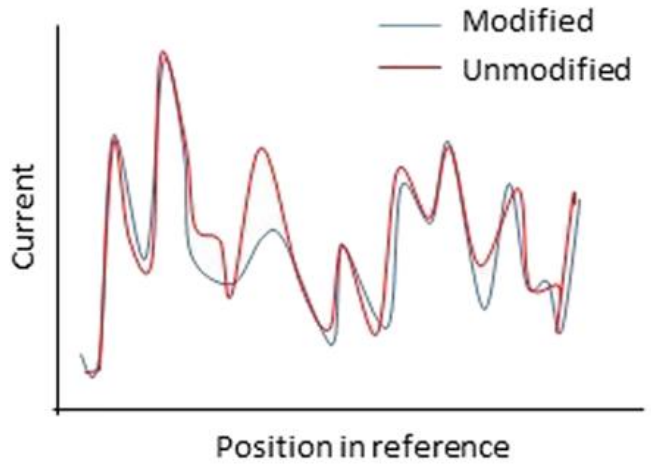


https://nanoporetech.com/platform/technology

Oxford Nanopore sequencing



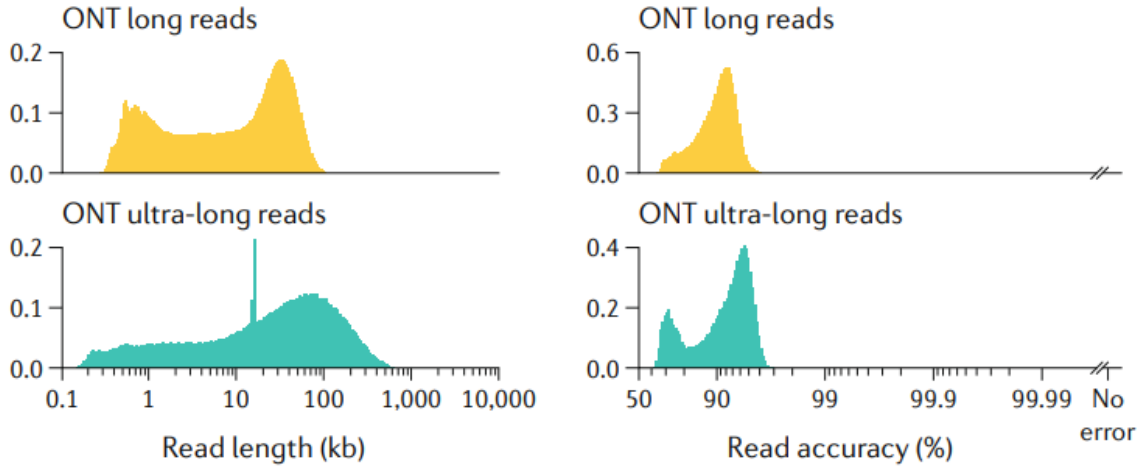
- Includes base modifications:
  - 5mC in DNA
  - m6A in RNA
  - Others with variable accuracy



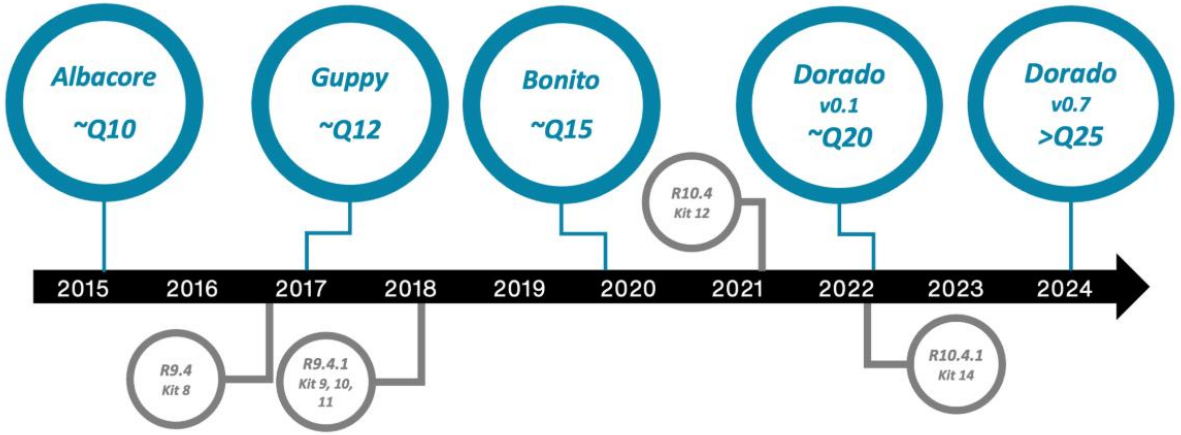
PMID: 31602005

Oxford Nanopore read profile

- Single end
- Very long reads (10 kb – 1M+)
- Higher error rate **especially at homopolymers**
- Captures base modifications



PMID: 32504078



https://nanoporetech.com/platform/technology

Oxford Nanopore summary

- Very long reads
- Lower accuracy
- Minimal prep, highly flexible



PromethION



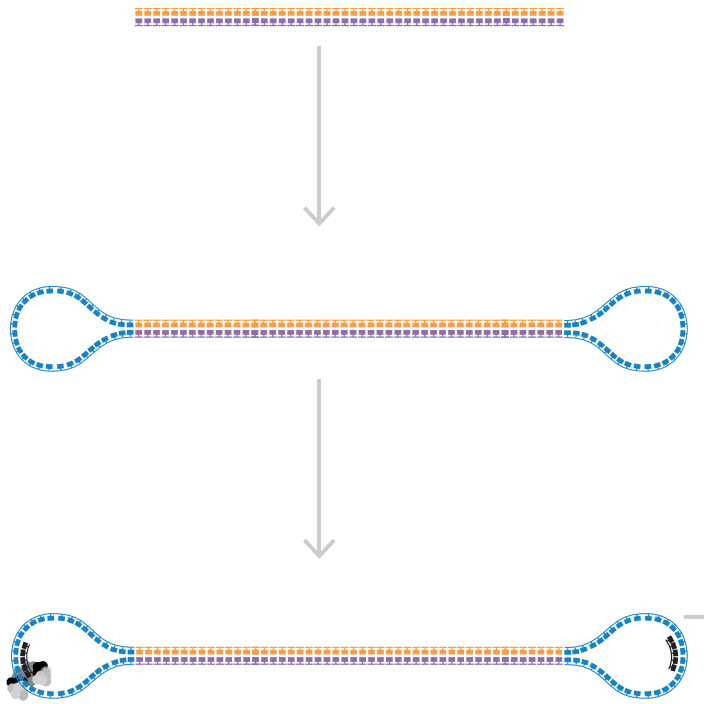
MinION

Pacific Biosciences HiFi sequencing

Start with high-quality double stranded DNA

Prepare SMRTbell libraries

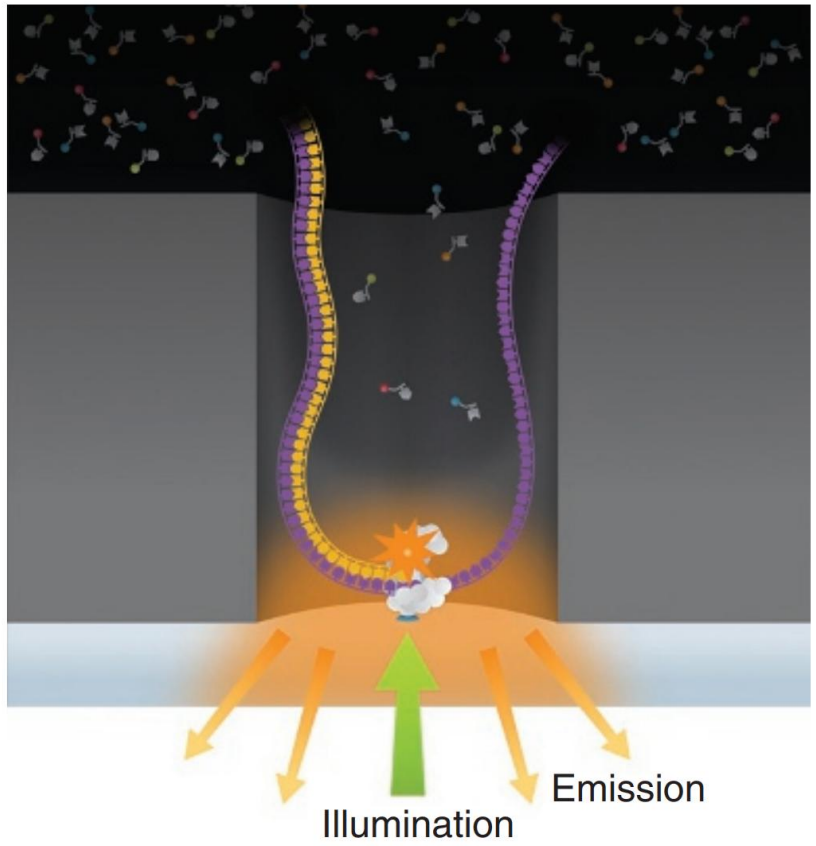
Anneal primers and bind DNA polymerase



- Create circular template

https://www.pacb.com/technology/hifi-sequencing/how-it-works/

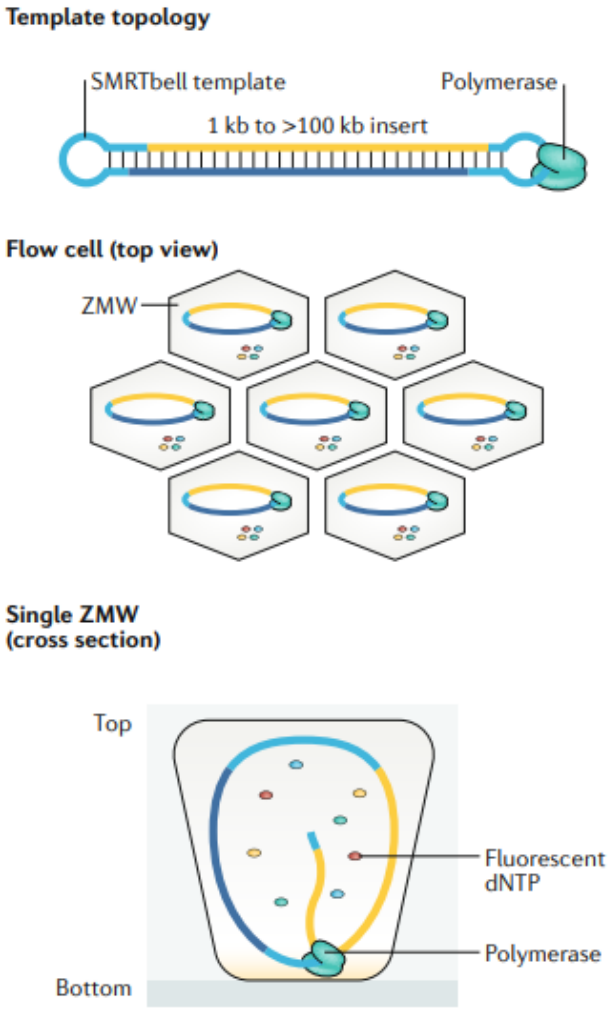
Pacific Biosciences HiFi sequencing



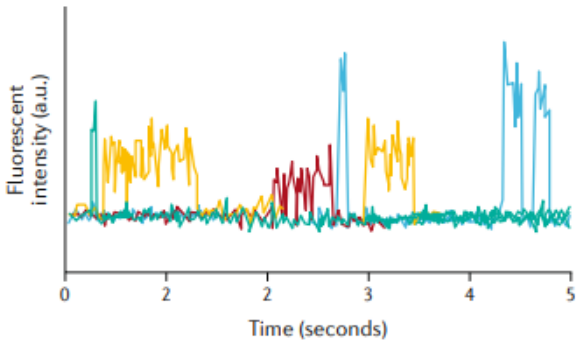
- Millions of nanowells (Zero-Mode Waveguide, ZMW)
- One polymerase each
- Sequencing by synthesis

PMID: 32504078

Pacific Biosciences HiFi sequencing

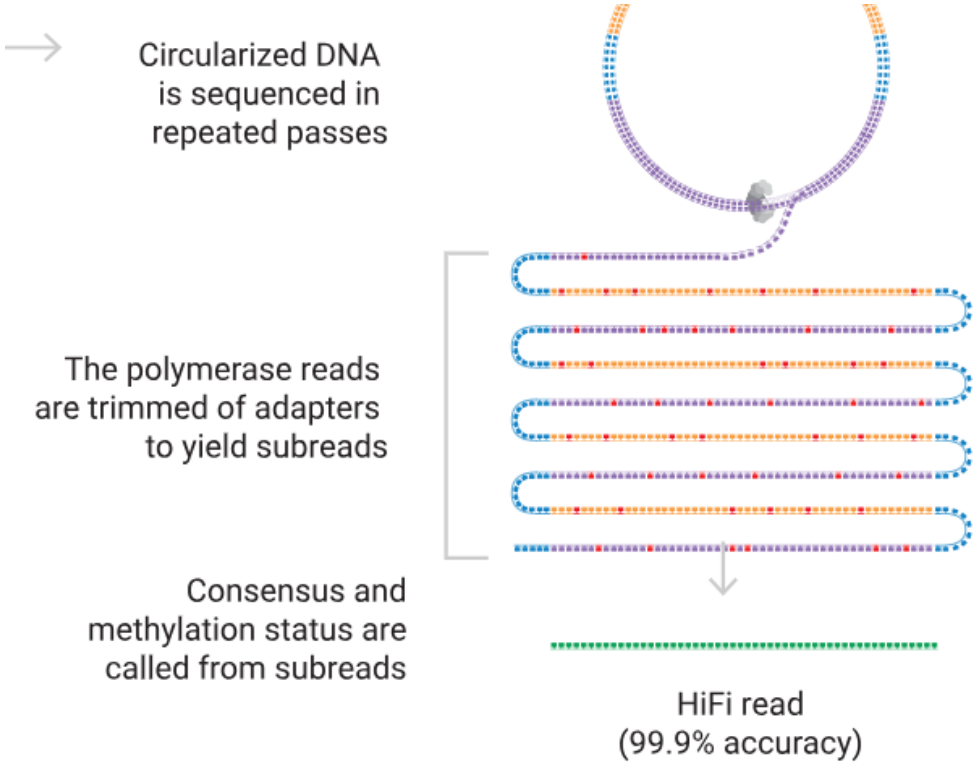


- Create circular template
- Millions of nanowells (Zero-Mode Waveguide, ZMW)
- One polymerase each
- Sequencing by synthesis
- Detect light emitted in nucleotide incorporation



PMID: 32504078

Pacific Biosciences HiFi sequencing

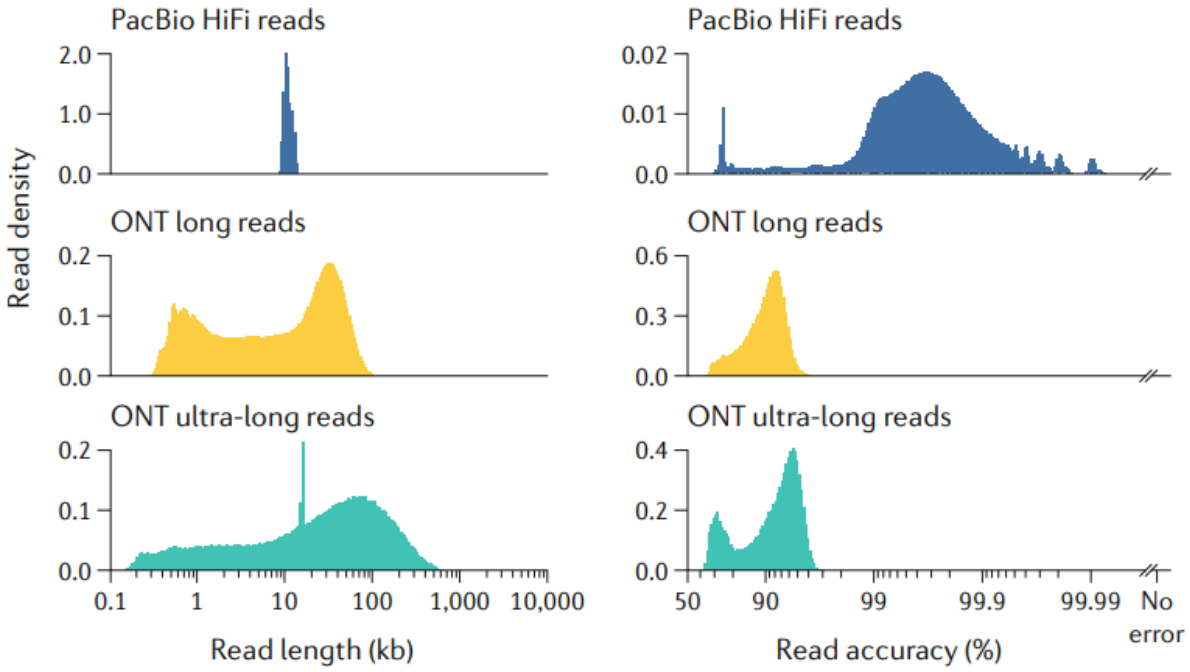


- Sequence repeatedly around circle and take consensus
- Individual accuracy < 90%
- Consensus accuracy > 99.9%

https://www.pacb.com/technology/hifi-sequencing/how-it-works/

PacBio read profile

- Single end
- Long reads (> 10 kb)
- Extremely low error rate
- Captures base modifications



PacBio summary

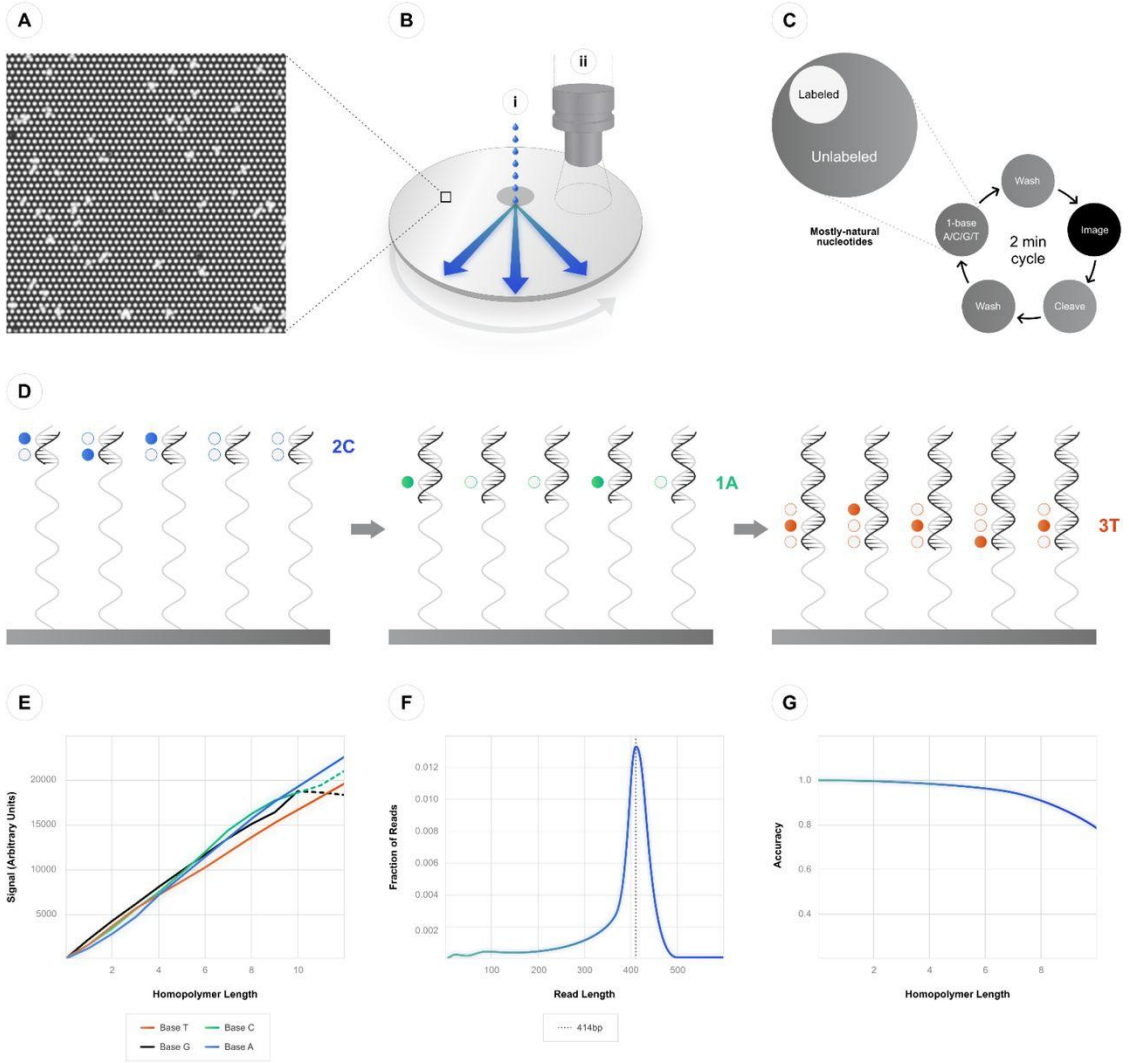
- Very high accuracy, long reads
- Widely used for genome assembly
- Higher cost per base
- Large footprint



Revio

Newer technologies: Ultima

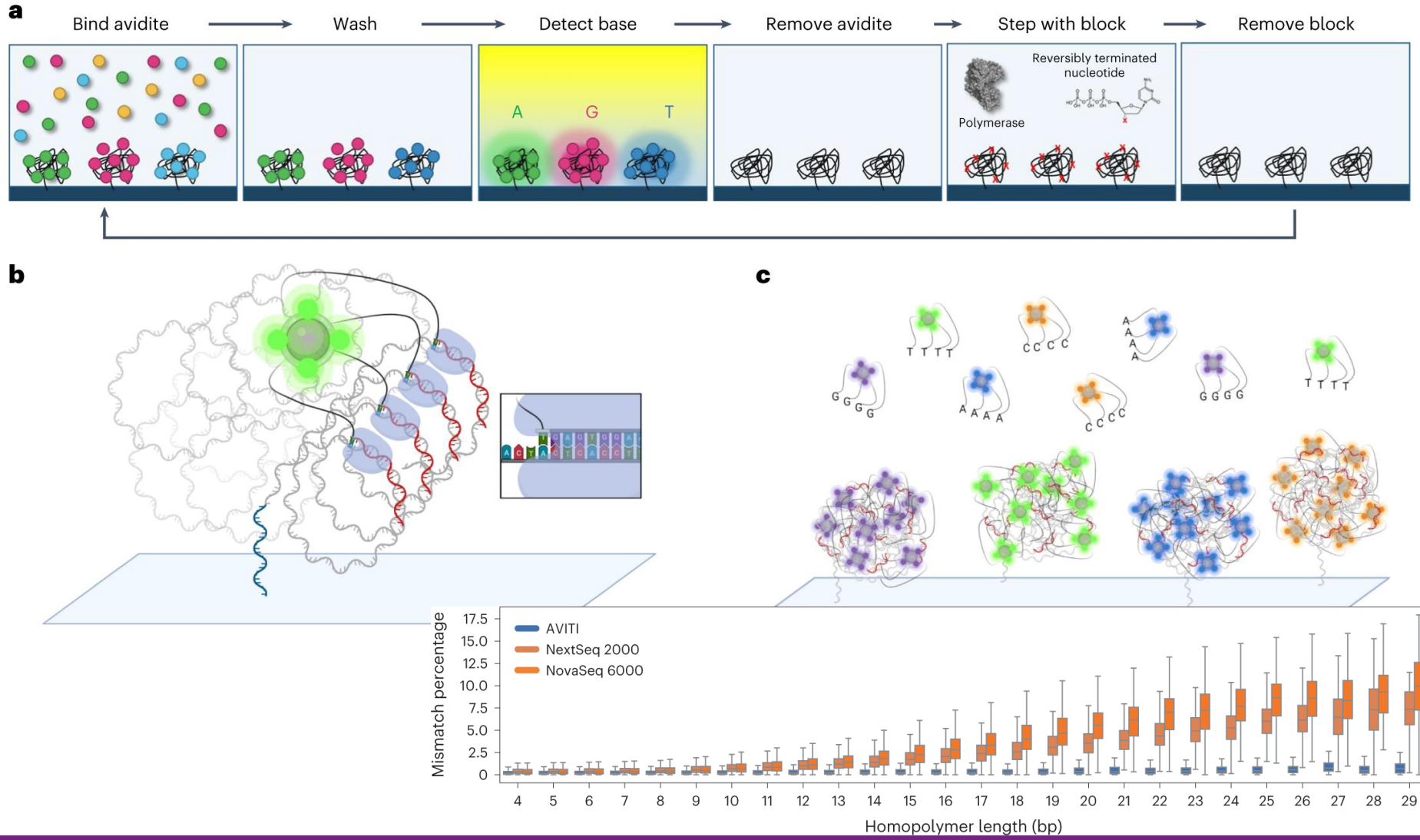
- Short read sequencing by synthesis
- Wafer chip
- Reduced use of expensive labelled dNTPs
- Recently released paired reads
- Pushing for $80 whole genome



https://www.biorxiv.org/content/10.1101/2022.05.29.493900v4.full

Newer technologies: AVITI

- Polymerase based
- Multivalent nucleotide binding
- Better homopolymer accuracy
- Paired short reads
- Not cheaper



Sequencing formats

- FASTA: Base sequences

```
>chromosome1
AGTCGATCGATCAGACTAACGATCGATCGAATCGAAT
```

- FASTQ: Base sequences and quality scores

```
>chromosome1
AGTCGATCGATCAGACTAACGATCGATCGAATCGAAT
+
CFHIGIGIJIIJFHIJJIIJIHFFFEDDDDDDDDDDD
```

- BAM/CRAM: Alignment formats (compared to a reference)

Sequencing quality

- Quality is measured by the Phred score (named after software Phred, used for Sanger sequencing analysis in the 1990s). Eg a “Q40 base” is highly accurate

$$Q = -10log10P_{error}$$

P_{error}(obs. base)	Base Quality	ASCII Character
50 %	3	$
32 %	5	&
10 %	10	+
1 %	20	5
0.1 %	30	?
0.01 %	40	I

Representing aligned reads

- Mapping/aligning reads to a reference genome is a core component of many analysis workflows. You'll hear more about this in later Modules

Mapping describes where in the reference genome a read matches. This read maps to position 1,020 on chr10

chr10:1,020 ATGAGAGAGTATCTCGACTCTAGGCCGATACCA

|||| ||||||||||

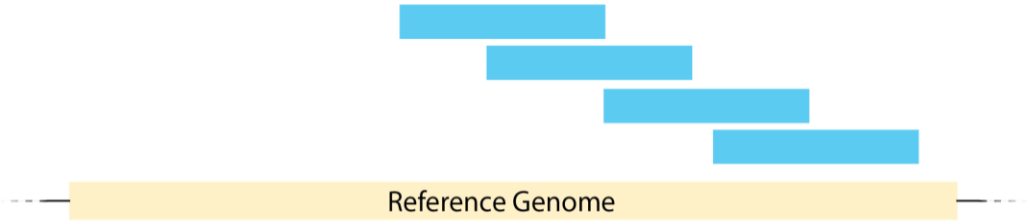
read-a AGTAACTCGACTCTA

Alignment describes how the bases in the read line up against the reference genome at the mapped location. This read has 14 matches and 1 mismatch.

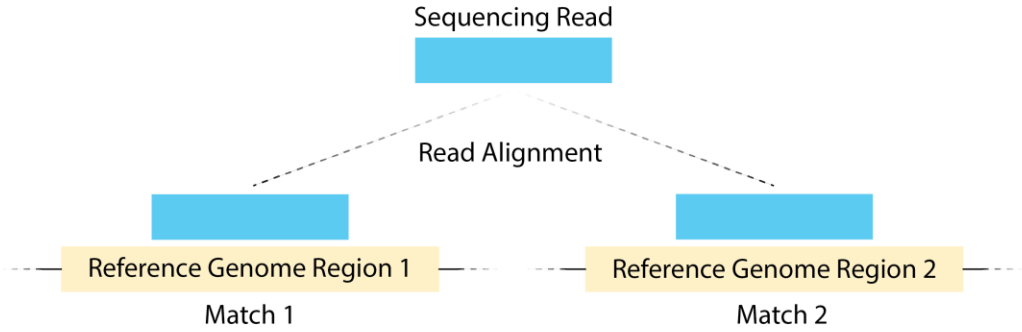
The location of the read on the reference genome and pattern of mismatches/indels is represented in bam/cram format.

Mapping quality

- Mapping quality is dependent on the **similarity of the match** and the **uniqueness of the match**
- More likely to be low in **repetitive regions**
- Also uses a Phred score



Single mapping location →
Good mapping quality, if few mismatches



Multi-mapped read →
Poor mapping quality (MQ0)

https://hbctraining.github.io/

We have a lot of tools















Coffee Break & Networking Session

Workshop Sponsors